

HS 224: Game Theory and Economics

BTech 4th Semester, Quiz 2

Total Marks 10, Time 55 minutes

Date: 4th April, 2018

[Answers should be accompanied by proper elaboration]

1. [Voting game with vote bank] Recall the setting of the electoral competition game. Suppose the distribution of favourite positions of voters has median at m . There are two candidates, 1 and 2.
 - (i) 20% of the extreme right-wing voters vote *in block* for the candidate whose position is more to the right. In case both candidates take the same position, the votes of the block get evenly distributed. The rest 80% of voters vote individually like in the original setting. Find the Nash equilibrium/equilibria. [1]
 - (ii) The block votes for 1 only, irrespective of the position 1 takes. Identify any one Nash equilibrium in this case. Can 2 win in the Nash equilibria? [2]
 - (iii) The block votes for 1. But if 1 takes a left-wing position the block does not vote for anyone. Specifically, if 1 takes a position $x_1 < m - k$ the block does not vote. Here k is a small number and 45% of voters have their favourite positions equal or less than $m - k$. Identify any one Nash equilibrium in this case. [2]
 - (iv) Between (ii) and (iii) how do the Nash equilibria differ? [2]
2. In the first price sealed-bid auction game prove that the winning bid always lies between v_1 and v_2 in the Nash equilibria. [3]

(x_1, m)
60% 40% 40%.